

TrES-3b

R-1512

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_190427 · created 2026-08-01T16:03:42+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458600.8233 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.125 +/- 0.032
Transit depth (Rp/Rs)^2	1.57 +/- 0.81 [%]
Orbital Inclination (inc)	83.36 +/- 1.7
Airmass coefficient 1 (a1)	1.188 +/- 0.015
Airmass coefficient 2 (a2)	-0.1405 +/- 0.0084
Scatter in the residuals of the lightcurve fit is	1.24 %
Best Comparison Star	None
Optimal Aperture	3.75
Optimal Annulus	8.2
Transit Duration (day)	0.06 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.125 ± 0.032 vs 0.16309 (deviation 1.19σ)
Duration fit vs published	0.06 d vs 0.0567 d (deviation 0.22σ)
Mid-time O–C	2.9 min
Depth SNR	3.9
Residual scatter	1.24 %

Light curve

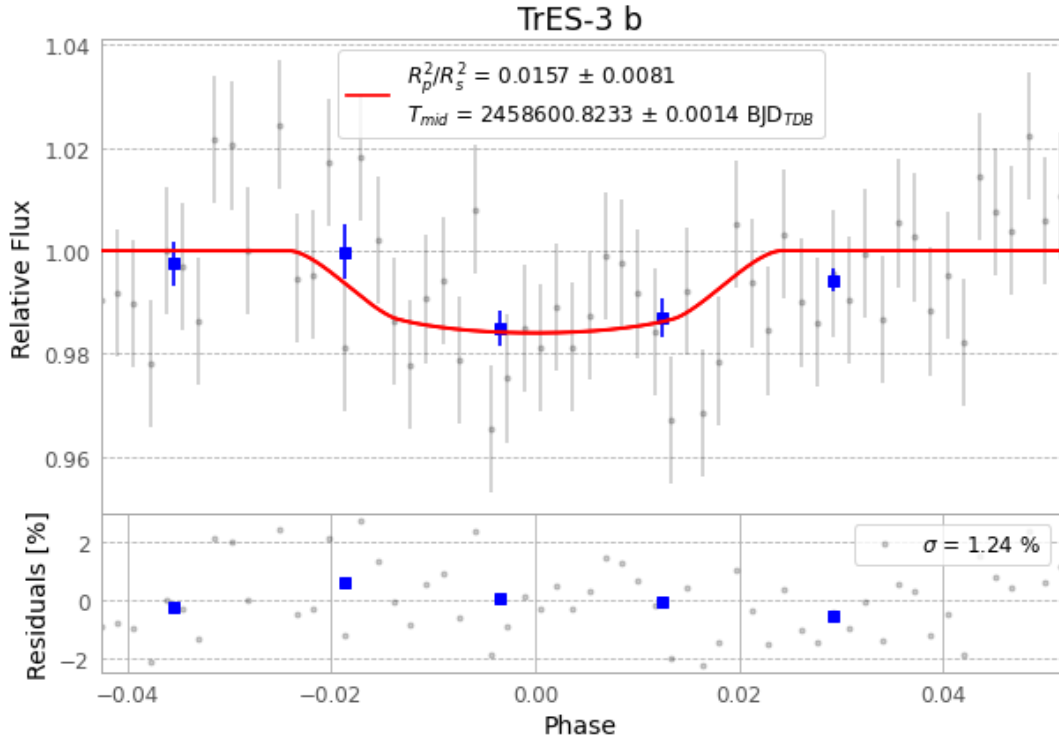


Figure 1 — FinalLightCurve_TrES-3 b_2019-04-26.png (SHA-256 4061c41a8e866b54...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [384, 74], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8d7434d299475630...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

62 FITS frames (41 MB), TRES-3190427061512.FITS → TRES-3190427091812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-190427.log (f177596242b1...), FinalLightCurve_TrES-3 b_2019-04-26.png (4061c41a8e86...), FinalLightCurve_TrES-3 b_2019-04-26.csv (e2b207d15707...)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 16:03Z.